

1

Growth hormone

COMP

You should be able to: Describe the direct and insulin like growth factor mediated actions of growth hormone and predict the result of excess or deficiency before and after epiphyseal closure.

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2

Thyroid hormone

COMP

You should be able to: Trace thyroid hormone synthesis, state the metabolic actions of T3 and T4, and match hyperthyroid and hypothyroid signs to the underlying rate of metabolism.

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3

COMP

Cortisol and the stress response

You should be able to: Describe the metabolic, immune, and cardiovascular actions of cortisol and explain the consequences of chronic elevation.

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4

COMP

Pancreatic islet hormones

You should be able to: Compare insulin and glucagon by stimulus and by effect on liver glycogen and on plasma glucose and identify the islet cell that secretes each.

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5**Diabetes mellitus**

COMP

You should be able to: Distinguish type 1 from type 2 diabetes by mechanism and explain how untreated hyperglycemia produces polyuria and ketoacidosis.

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6

COMP

Glucose tolerance testing

You should be able to: Plot a glucose tolerance curve from measured or simulated data and distinguish a normal response from an impaired one.

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7

COMP

Four digestive processes

You should be able to: Distinguish motility, secretion, digestion, and absorption and identify where in the tract each dominates.

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8

COMP

Motility patterns

You should be able to: Compare peristalsis, segmentation, and the migrating motor complex by pattern and purpose.

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COMP

Neural and hormonal regulation

You should be able to: Explain short and long reflex control by the enteric nervous system and match gastrin, secretin, cholecystokinin, and glucose dependent insulinotropic peptide to their stimulus and action.

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10

COMP

Phases of digestive control

You should be able to: Describe the cephalic, gastric, and intestinal phases and state the dominant control mechanism in each.

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11

COMP

Gastric secretion and mucosal protection

You should be able to: Trace the production of hydrochloric acid by the parietal cell and explain how the mucosal barrier protects the stomach and how ulcers develop when it fails.

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COMP

Pancreatic and biliary secretion

You should be able to: State the components of pancreatic juice and bile and explain how bicarbonate neutralizes chyme and how bile salts emulsify fat.

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13

COMP

Carbohydrate and protein digestion and absorption

You should be able to: Trace a starch and a protein from mouth to bloodstream naming the enzymes at each step and the transport mechanism at the brush border.

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COMP

Lipid digestion and absorption

You should be able to: Trace a triglyceride through emulsification, lipase action, micelle formation, and chylomicron packaging to the lymphatic route and explain why fat takes a different path than glucose.

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COMP

Liver function and enterohepatic circulation

You should be able to: State the metabolic, storage, and detoxification functions of the liver and trace the enterohepatic circulation of bile salts.

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COMP

Large intestine and microbiome

You should be able to: Describe water and electrolyte absorption in the colon, the contribution of the microbiome, and the defecation reflex.

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COMP

Digestive enzyme experiment

You should be able to: Test the effect of pH, temperature, and substrate on a digestive enzyme and relate the results to the region of the tract where that enzyme works.

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COMP

Absorptive state

You should be able to: Describe the fate of glucose, amino acids, and fats in the absorptive state and identify insulin as the dominant hormone.

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COMP

Postabsorptive state

You should be able to: Explain how glycogenolysis, gluconeogenesis, lipolysis, and ketogenesis maintain plasma glucose during fasting and name the hormones that drive each.

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20

COMP

Integrated glucose regulation

You should be able to: Predict the hormonal and metabolic response to a carbohydrate meal and to a prolonged fast and explain how plasma glucose stays within range in both.

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COMP

Energy balance and metabolic rate

You should be able to: Define basal metabolic rate, state the factors that raise and lower it, and explain how indirect calorimetry estimates energy expenditure.

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COMP

Thermoregulation

You should be able to: Trace the negative feedback response to heat and to cold through the hypothalamus and the effectors and explain the mechanism of fever.

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COMP

Metabolic rate measurement

You should be able to: Estimate metabolic rate from measured or simulated oxygen consumption and compare resting values with values after activity.

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